

TMUA Formula + Fact Sheet

A compact but specification-aware reference sheet for TMUA Paper 1 and Paper 2. Use it for recall, but practise applying these under timed conditions.

Exam reminder: There is no formula booklet in TMUA. Paper 1 tests mathematical application. Paper 2 tests mathematical reasoning and proof, using Section 1 knowledge plus logic and argument skills. There is no negative marking, so attempt every question.

1. Algebra, indices, surds and polynomials

INDICES

Index laws: $a^m a^n = a^{m+n}$, $\frac{a^m}{a^n} = a^{m-n}$

$$(a^m)^n = a^{mn}, (ab)^n = a^n b^n, a^{-n} = \frac{1}{a^n}$$

$$a^{1/n} = \sqrt[n]{a}, a^{m/n} = \sqrt[n]{a^m}$$

TMUA uses rational, negative and fractional indices frequently.

SURDS

$$\sqrt{ab} = \sqrt{a}\sqrt{b}, \sqrt{\frac{a}{b}} = \frac{\sqrt{a}}{\sqrt{b}}$$

$$\sqrt{a^2 b} = a\sqrt{b} \text{ for } a \geq 0$$

$$\text{Conjugates: } (a + \sqrt{b})(a - \sqrt{b}) = a^2 - b$$

$$\frac{1}{a + \sqrt{b}} = \frac{a - \sqrt{b}}{a^2 - b}$$

Always simplify surds and rationalise denominators.

QUADRATICS

For $ax^2 + bx + c = 0$:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Discriminant: $\Delta = b^2 - 4ac$

$\Delta > 0$: two real roots; $\Delta = 0$: repeated root; $\Delta < 0$: no real roots.

COMPLETING THE SQUARE

$$ax^2 + bx + c = a\left(x + \frac{b}{2a}\right)^2 + c - \frac{b^2}{4a}$$

For $x^2 + bx + c$:

$$x^2 + bx + c = \left(x + \frac{b}{2}\right)^2 + c - \frac{b^2}{4}$$

Use for minimum/maximum value, vertex and inequalities.

ROOTS

If $ax^2 + bx + c = 0$ has roots α, β :

$$\alpha + \beta = -\frac{b}{a}, \quad \alpha\beta = \frac{c}{a}$$

Vertex of $y = ax^2 + bx + c$:

$$x = -\frac{b}{2a}$$

FACTORISATIONS

$$a^2 - b^2 = (a - b)(a + b)$$

$$a^3 + b^3 = (a + b)(a^2 - ab + b^2)$$

$$a^3 - b^3 = (a - b)(a^2 + ab + b^2)$$

$$x^2 + bx + c = (x + p)(x + q) \text{ if } p + q = b, pq = c$$

FACTOR AND REMAINDER THEOREM

Factor theorem: $f(a) = 0 \iff (x - a)$ is a factor of $f(x)$.

Remainder theorem: remainder on division by $(x - a)$ is $f(a)$.

For divisor $ax + b$, set $ax + b = 0$, so remainder is $f(-b/a)$.

ALGEBRAIC DIVISION AND IDENTITIES

If $f(x) = (x - a)q(x) + r$, then $r = f(a)$.

An **identity** is true for all allowed values of x .

An **equation** is true only for its solutions.

Compare coefficients when two polynomial expressions are identical.

SIMULTANEOUS EQUATIONS AND INTERSECTIONS

Line/quadratic method: substitute the linear expression into the quadratic, solve the resulting quadratic, then back-substitute.

Geometric meaning: solutions are intersection points of the two graphs.

A line can meet a quadratic in 0, 1, or 2 points; use the discriminant after substitution.

QUADRATIC SHORTCUTS

For $f(x) = ax^2 + bx + c$, with a non-zero coefficient:

$$f(x) > 0 \text{ for all real } x \iff a > 0 \text{ and } \Delta < 0$$

$$f(x) \geq 0 \text{ for all real } x \iff a > 0 \text{ and } \Delta \leq 0$$

Always negative: use $a < 0$ with the same discriminant conditions.

Tangency after substitution: $\Delta = 0$ (one repeated intersection).

$$\text{If roots are } \alpha, \beta: \quad \alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$$

$$1/\alpha + 1/\beta = (\alpha + \beta)/(\alpha\beta)$$

2. Inequalities, functions and graphs

INEQUALITY RULES

If $a < b$, then $a + c < b + c$.

If $c > 0$, then $ac < bc$.

If $c < 0$, then $ac > bc$. **Reverse the sign.**

For products/quotients, use critical values and a sign chart.

MODULUS

$|x| < a \iff -a < x < a$, for $a > 0$

$|x| > a \iff x < -a$ or $x > a$, for $a > 0$

$|x - a|$ means distance from a on the number line.

FUNCTIONS

$f(g(x))$ means apply g first, then f .

$\sqrt{x}$ means the **positive** square root.

One-to-one: each output comes from at most one input.

Many-to-one: different inputs can give the same output.

QUADRATIC INEQUALITIES

To solve $f(x) > 0$, find the roots and identify where the graph is above the x -axis.

For $a(x - r_1)(x - r_2)$: outside the roots if $a > 0$; inside the roots if $a < 0$.

Sketch first. It prevents most sign errors.

AM-GM

For $a, b \geq 0$:

$$\frac{a+b}{2} \geq \sqrt{ab}$$

Equality holds iff $a = b$.

Useful for minimum-value problems such as $x + \frac{1}{x}$.

COMMON GRAPHS

- Line: $y = mx + c$
- Quadratic: $y = ax^2 + bx + c$
- Cubic: possible one, two, or three real roots
- Reciprocal: $y = \frac{1}{x}$, asymptotes $x = 0, y = 0$
- Exponential: $y = a^x, a > 0$
- Logarithm: $y = \log_a x, x > 0$
- Modulus: $y = |x|$

GRAPH TRANSFORMATIONS

New graph	Transformation from $y = f(x)$
$y = f(x) + a$	up a
$y = f(x) - a$	down a
$y = f(x + a)$	left a
$y = f(x - a)$	right a
$y = af(x)$	vertical stretch by factor a
$y = f(ax)$	horizontal stretch by factor $\frac{1}{a}$
$y = -f(x)$	reflection in the x -axis
$y = f(-x)$	reflection in the y -axis

LOGARITHMS

$$a^b = c \iff b = \log_a c$$

$$\log_a(xy) = \log_a x + \log_a y$$

$$\log_a(x/y) = \log_a x - \log_a y$$

$$\log_a(x^k) = k \log_a x$$

$$\log_a(1/x) = -\log_a x, \log_a a = 1$$

EXPONENTIALS

$$y = a^x \iff x = \log_a y$$

$$\ln x = \log_e x$$

$$e^{\ln x} = x, \ln(e^x) = x$$

extra Change of base: $\log_a x = \frac{\log_b x}{\log_b a}$

Change of base is useful but not required by the 2026 spec.

DOMAIN + INVERSE QUICK CHECK

For a real-valued expression: denominator $\neq 0$; even-root radicand ≥ 0 ; log argument > 0 .

For $f(g(x))$: x must be in $\text{Dom}(g)$, and $g(x)$ must lie in $\text{Dom}(f)$.

Inverse function: write $y = f(x)$, swap x and y , then solve for y .

$$\text{Dom}(f^{-1}) = \text{Range}(f), \quad \text{Range}(f^{-1}) = \text{Dom}(f)$$

Graphs of f and f^{-1} reflect in $y = x$.

3. Sequences, series, binomial and counting

ARITHMETIC SEQUENCES

$$u_n = a + (n-1)d$$

$$S_n = \frac{n}{2}[2a + (n-1)d] = \frac{n}{2}(a+l)$$

First difference is constant.

GEOMETRIC SEQUENCES

$$u_n = ar^{n-1}$$

Finite sum:

$$S_n = \frac{a(1-r^n)}{1-r}, \quad r \neq 1$$

Infinite sum:

$$S_\infty = \frac{a}{1-r}, \quad |r| < 1$$

STANDARD SUMS

$$\sum_{k=1}^n k = \frac{n(n+1)}{2}$$

$$\sum_{k=1}^n c = cn$$

Useful in arithmetic series and proof-style questions.

RECURRENCE RELATIONS

A recurrence defines terms using previous terms, e.g.

$$x_{n+1} = f(x_n)$$

Generate terms carefully: start from the given x_1 or x_0 .

LINEAR AND QUADRATIC SEQUENCES

Linear sequence: constant first difference.

Quadratic sequence: constant second difference.

If $u_n = an^2 + bn + c$, then second difference = $2a$.

BINOMIAL EXPANSION

$$(a+b)^n = \sum_{r=0}^n \binom{n}{r} a^{n-r} b^r$$

$$\binom{n}{r} = \frac{n!}{r!(n-r)!} = \binom{n}{n-r}$$

For coefficient of x^k , find the term where powers combine to x^k .

COMBINATIONS

$${}^nC_r = \binom{n}{r} = \frac{n!}{r!(n-r)!}$$

Use when order does **not** matter.

PERMUTATIONS

$${}^nP_r = \frac{n!}{(n-r)!}$$

Use when order **does** matter.

SYSTEMATIC COUNTING

If one task has m choices and the next has n choices, the ordered pair of tasks has mn choices.

Use lists, grids, tables and tree diagrams to avoid double-counting.

For repeated choices, check carefully whether repetition is allowed.

SEQUENCE SHORTCUTS

If S_n is the sum of the first n terms:

$$u_n = S_n - S_{n-1} \quad (n \geq 2)$$

Three consecutive terms a, b, c :

$$\text{AP: } 2b = a + c$$

$$\text{GP: } b^2 = ac$$

These relations often remove the need to introduce d or r explicitly.

BINOMIAL SHORTCUTS

$$C(n,r) = C(n,n-r)$$

$$\sum C(n,r) = 2^n$$

For a polynomial $P(x)$: sum of coefficients = $P(1)$; alternating sum = $P(-1)$.

Before expanding, try $x = 1$ or $x = -1$; it can collapse a long coefficient question instantly.

4. Coordinate geometry and circle geometry

STRAIGHT LINES

Gradient:

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

Point-gradient form:

$$y - y_1 = m(x - x_1)$$

General form: $ax + by + c = 0$

Parallel: $m_1 = m_2$

Perpendicular: $m_1 m_2 = -1$

DISTANCE AND MIDPOINT

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

CIRCLES: EQUATIONS

$$(x - a)^2 + (y - b)^2 = r^2$$

Centre (a, b) , radius r .

If $x^2 + y^2 + 2gx + 2fy + c = 0$:

$$\text{centre} = (-g, -f), \quad r = \sqrt{g^2 + f^2 - c}$$

TANGENT TO CIRCLE

The tangent at a point is perpendicular to the radius at that point.

If radius gradient is m , tangent gradient is $-1/m$.

Find the radius gradient first, then use perpendicular-gradient rule.

CIRCLE THEOREMS

- Angle at the centre is twice the angle at the circumference.
- Angle in a semicircle is 90° .
- Angles in the same segment are equal.
- Opposite angles in a cyclic quadrilateral sum to 180° .
- Tangent is perpendicular to radius at the point of contact.
- Angle between tangent and chord equals the angle in the alternate segment.
- Perpendicular from centre to chord bisects the chord.

ANGLE FACTS

Angles on a straight line sum to 180° .

Angles around a point sum to 360° .

Vertically opposite angles are equal.

Triangle angles sum to 180° .

Exterior angle of triangle = sum of opposite interior angles.

PARALLEL LINES

Corresponding angles are equal.

Alternate angles are equal.

Co-interior angles sum to 180° .

POLYGONS

Sum of interior angles of n -gon:

$$(n - 2)180^\circ$$

Each interior angle of regular n -gon:

$$\frac{(n - 2)180^\circ}{n}$$

Sum of exterior angles: 360°

PYTHAGORAS

$$a^2 + b^2 = c^2$$

In 3D cuboid diagonal:

$$d = \sqrt{l^2 + w^2 + h^2}$$

LINE + COORDINATE SHORTCUTS

Point $P(x_0, y_0)$ to $Ax + By + C = 0$:

$$d = |Ax_0 + By_0 + C| / \sqrt{A^2 + B^2}$$

Angle between gradients m_1, m_2 :

$$\tan \theta = |(m_2 - m_1) / (1 + m_1 m_2)|$$

Triangle area from coordinates:

$$A = \frac{1}{2} |x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)|$$

$A = 0$ iff the three points are collinear.

CIRCLE SHORTCUTS

Let d be the perpendicular distance from the centre of a circle of radius r to a line:

$d < r$: 2 intersections

$d = r$: tangent

$d > r$: no intersection

Chord at distance d from centre:

$$\text{chord length} = 2\sqrt{r^2 - d^2}$$

5. Trigonometry, radians and geometry measurement

RIGHT-TRIANGLE TRIG

$$\sin \theta = \frac{\text{opposite}}{\text{hypotenuse}}$$

$$\cos \theta = \frac{\text{adjacent}}{\text{hypotenuse}}$$

$$\tan \theta = \frac{\text{opposite}}{\text{adjacent}}$$

CORE IDENTITIES

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$1 + \tan^2 \theta = \sec^2 \theta$$

DOUBLE ANGLE

$$\sin 2\theta = 2 \sin \theta \cos \theta$$

$$\cos 2\theta = \cos^2 \theta - \sin^2 \theta$$

$$\cos 2\theta = 1 - 2 \sin^2 \theta = 2 \cos^2 \theta - 1$$

SINE AND COSINE RULES

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$a^2 = b^2 + c^2 - 2bc \cos A$$

Area:

$$\text{Area} = \frac{1}{2} ab \sin C$$

Remember the ambiguous SSA case for sine rule.

EXACT VALUES

θ	0°	30°	45°	60°	90°
$\sin \theta$	0	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	1
$\cos \theta$	1	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$	0
$\tan \theta$	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	undefined

RADIANS

$$180^\circ = \pi \text{ radians}$$

$$\theta^\circ = \frac{\pi \theta}{180} \text{ radians}$$

$$\theta \text{ radians} = \frac{180\theta^\circ}{\pi}$$

ARCS AND SECTORS

For θ in radians:

$$\text{arc length} = r\theta$$

$$\text{sector area} = \frac{1}{2} r^2 \theta$$

$$\text{segment area} = \text{sector area} - \text{triangle area}$$

TRIG GRAPHS

$\sin x, \cos x$: period 360° or 2π
 $\tan x$: period 180° or π
 $\sin(-x) = -\sin x, \cos(-x) = \cos x$

BEARINGS

Bearings are measured clockwise from North.
 Always written with three figures, e.g. $047^\circ, 230^\circ$.
 Draw North lines at every point involved.

BASIC AREAS

Triangle: $\frac{1}{2}bh$
 Parallelogram: bh
 Trapezium: $\frac{1}{2}(a+b)h$
 Circle: πr^2
 Circumference: $2\pi r = \pi d$

VOLUMES

Cuboid: lwh
 Prism: cross-sectional area $\times$ length
 Cylinder: $\pi r^2 h$
 Sphere, cone and pyramid formulae are usually given if needed.

SOLVING TRIG EQUATIONS

1. Rearrange to $\sin u = k$, $\cos u = k$, or $\tan u = k$. Find the reference angle and all solutions in the interval.
2. If $u = kx + c$, transform the x-interval into the u-interval before solving.
 Example: $\sin 2x = 1/2$ with x in $[0, \pi]$: solve $\sin u = 1/2$ for u in $[0, 2\pi]$, then divide u by 2.
3. When useful, replace $\sin^2 x$ by $1 - \cos^2 x$ (or $\cos^2 x$ by $1 - \sin^2 x$) to get a quadratic in one trig function.
4. Use graph/quadrant symmetry to keep every valid root; do not stop after the principal value.
5. Always check the original equation and the original x-interval.

6. Differentiation and integration

DERIVATIVE MEANING

$$f'(x) = \frac{dy}{dx}$$

Derivative = gradient of tangent = instantaneous rate of change.

Second derivative: $f''(x) = \frac{d^2y}{dx^2}$

POWER RULE

$$\frac{d}{dx}(x^n) = nx^{n-1}$$

Works for rational n , where expression is defined.

Simplify first, then differentiate.

TANGENTS AND NORMALS

Tangent gradient at $x = a$: $m = f'(a)$

Tangent:

$$y - y_1 = m(x - x_1)$$

Normal gradient:

$$m_{\text{normal}} = -\frac{1}{m}$$

STATIONARY POINTS

Stationary point: $f'(x) = 0$

If $f''(x) > 0$: local minimum.

If $f''(x) < 0$: local maximum.

If $f''(x) = 0$: inspect further.

INCREASING/DECREASING

If $f'(x) > 0$, then f is increasing.

If $f'(x) < 0$, then f is decreasing.

Use sign charts for intervals.

EXTRA DERIVATIVE RULES

extra Product: $(uv)' = u'v + uv'$

extra Quotient:

$$\left(\frac{u}{v}\right)' = \frac{u'v - uv'}{v^2}$$

extra Chain:

$$\frac{d}{dx}f(g(x)) = f'(g(x))g'(x)$$

EXTRA STANDARD DERIVATIVES

extra $\frac{d}{dx}e^x = e^x$

extra $\frac{d}{dx}\ln x = \frac{1}{x}$

extra $\frac{d}{dx}\sin x = \cos x$

extra $\frac{d}{dx}\cos x = -\sin x$

POWER INTEGRALS

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C, \quad n \neq -1$$

For definite integrals:

$$\int_a^b f(x) dx = F(b) - F(a)$$

EXTRA STANDARD INTEGRALS

extra $\int \frac{1}{x} dx = \ln|x| + C$

extra $\int e^x dx = e^x + C$

extra $\int \cos x dx = \sin x + C$

extra $\int \sin x dx = -\cos x + C$

FUNDAMENTAL THEOREM

If $F'(x) = f(x)$, then:

$$\int_a^b f(x) dx = F(b) - F(a)$$

$$\frac{d}{dx} \left(\int_a^x f(t) dt \right) = f(x)$$

COMBINING INTEGRALS

$$\int_a^b [f(x) + g(x)] dx = \int_a^b f(x) dx + \int_a^b g(x) dx$$

$$\int_a^b f(x) dx + \int_b^c f(x) dx = \int_a^c f(x) dx$$

$$\int_b^a f(x) dx = -\int_a^b f(x) dx$$

AREA VS INTEGRAL

Definite integral can be negative.

Area between curve and x -axis uses positive area.

If curve crosses the axis, split at roots.

TRAPEZIUM RULE

For equal spacing h :

$$A \approx \frac{h}{2} [y_0 + y_n + 2(y_1 + \dots + y_{n-1})]$$

Over/underestimate depends on curvature.

DIFFERENTIAL EQUATIONS

If $\frac{dy}{dx} = f(x)$, then:

$$y = \int f(x) dx$$

Use given point/condition to find C .

7. Vectors, transformations and similarity

COLUMN VECTORS

$$\begin{pmatrix} a \\ b \end{pmatrix} + \begin{pmatrix} c \\ d \end{pmatrix} = \begin{pmatrix} a+c \\ b+d \end{pmatrix}$$

$$k \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} ka \\ kb \end{pmatrix}$$

VECTOR BETWEEN POINTS

From $A(x_1, y_1)$ to $B(x_2, y_2)$:

$$\vec{AB} = \begin{pmatrix} x_2 - x_1 \\ y_2 - y_1 \end{pmatrix}$$

Magnitude:

$$\left| \begin{pmatrix} a \\ b \end{pmatrix} \right| = \sqrt{a^2 + b^2}$$

PARALLEL VECTORS

Vectors are parallel if one is a scalar multiple of the other:

$$\mathbf{a} = k\mathbf{b}$$

Collinear points often require parallel vectors.

VECTOR PROOFS

To prove a quadrilateral is a parallelogram, show opposite sides are equal and parallel as vectors.

To prove midpoint, show position vector is average of endpoints.

$$\mathbf{m} = \frac{\mathbf{a} + \mathbf{b}}{2}$$

SIMILARITY

Scale factor k :

Lengths scale by k .

Areas scale by k^2 .

Volumes scale by k^3 .

CONGRUENCE

Triangle congruence tests:

SSS, SAS, ASA, RHS.

Congruent shapes have equal corresponding lengths and angles.

TRANSFORMATIONS

- Translation: every point moves by the same vector.
- Reflection: mirror line is perpendicular bisector between point and image.
- Rotation: use centre, angle and direction.
- Enlargement: use centre and scale factor.
- Negative scale factor: enlargement plus reversal through centre.

QUADRILATERALS

- Parallelogram: opposite sides parallel and equal.
- Rectangle: parallelogram with right angles.
- Rhombus: all sides equal.
- Square: rectangle and rhombus.
- Kite: two pairs of adjacent equal sides.
- Trapezium: one pair of parallel sides.

3D GEOMETRY VOCABULARY

Know: face, surface, edge, vertex, prism, cylinder, cone, sphere, hemisphere, pyramid, plan and elevation.

For 3D Pythagoras, first find a diagonal on one face, then use it in a second right triangle.

8. Number, units, ratio, percentage and bounds

STANDARD FORM

$$a \times 10^n, \quad 1 \leq a < 10$$

For multiplication, multiply a 's and add powers of 10.

For division, divide a 's and subtract powers of 10.

UNITS

Speed:

$$\text{speed} = \frac{\text{distance}}{\text{time}}$$

Density:

$$\rho = \frac{m}{V}$$

Pressure:

$$P = \frac{F}{A}$$

Convert units before calculating.

PRIME FACTORISATION

Write numbers as products of prime powers.

HCF: take smaller powers common to both.

LCM: take larger powers appearing in either.

RATIO

For ratio $a : b$, total parts = $a + b$.

First share: $\frac{a}{a+b} \times \text{total}$

Second share: $\frac{b}{a+b} \times \text{total}$

Scale factor = $\frac{\text{new length}}{\text{old length}}$

DIRECT PROPORTION

$$y \propto x \iff y = kx$$

$$y \propto x^n \iff y = kx^n$$

Find k using one known pair.

INVERSE PROPORTION

$$y \propto \frac{1}{x} \iff y = \frac{k}{x}$$

$$y \propto \frac{1}{x^n} \iff y = \frac{k}{x^n}$$

PERCENTAGES

$$\% \text{ change} = \frac{\text{new} - \text{old}}{\text{old}} \times 100$$

Increase by $r\%$: multiply by $1 + \frac{r}{100}$.

Decrease by $r\%$: multiply by $1 - \frac{r}{100}$.

GROWTH AND DECAY

Compound growth:

$$A = P(1 + r)^n$$

Compound decay:

$$A = P(1 - r)^n$$

Here r is decimal rate, not percentage.

BOUNDS

If x is rounded to nearest unit:

$$x - \frac{1}{2} \leq \text{true value} < x + \frac{1}{2}$$

Nearest tenth: ± 0.05

Nearest hundredth: ± 0.005

RECURRING DECIMALS

Let x equal the recurring decimal.

Multiply by 10, 100, or 1000 to align recurring parts.

Subtract equations to remove recurrence.

APPROXIMATION

Use sensible estimates before calculating. In no-calculator TMUA, approximation often eliminates answer choices.

Always track whether the question asks for exact value, approximation, upper bound, lower bound, or error interval.

9. Probability and statistics

PROBABILITY SCALE

$$0 \leq P(A) \leq 1$$

Impossible: 0. Certain: 1.

Exhaustive outcomes sum to 1.

ADDITION RULE

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

If mutually exclusive:

$$P(A \cup B) = P(A) + P(B)$$

INDEPENDENCE

Independent iff:

$$P(A \cap B) = P(A)P(B)$$

Equivalently:

$$P(A | B) = P(A)$$

EXPECTED FREQUENCY

$$\text{expected frequency} = nP(A)$$

Relative frequency:

$$\frac{\text{number of successful outcomes}}{\text{number of trials}}$$

AVERAGES

$$\text{mean} = \frac{\sum x}{n}$$

$$\text{range} = \text{max} - \text{min}$$

Median = middle value after ordering.

Mode = most frequent value.

HISTOGRAMS

$$\text{frequency density} = \frac{\text{frequency}}{\text{class width}}$$

$$\text{frequency} = \text{frequency density} \times \text{class width}$$

SCATTER GRAPHS

Positive correlation: as one variable increases, the other tends to increase.

Negative correlation: as one variable increases, the other tends to decrease.

Correlation does not imply causation.

Interpolation is within the data range; extrapolation is outside and less reliable.

COMPLEMENT

$$P(A') = 1 - P(A)$$

Use "not" or "at least one" problems by complement.

CONDITIONAL PROBABILITY

$$P(A | B) = \frac{P(A \cap B)}{P(B)}, \quad P(B) > 0$$

"Given B" means restrict the sample space to B.

TREE DIAGRAMS

Multiply along branches.

Add separate routes to the same outcome.

Dependent events: second probability changes after first outcome.

Independent events: second probability stays the same.

VENN COUNTING

$$|A \cup B| = |A| + |B| - |A \cap B|$$

$$|A'| = |U| - |A|$$

Fill the intersection first.

GROUPED DATA

Estimated mean:

$$\bar{x} \approx \frac{\sum fx}{\sum f}$$

Use class midpoints for x.

Modal class = class with highest frequency.

QUARTILES AND SPREAD

$$IQR = Q_3 - Q_1$$

Use median and IQR to compare skewed data sets.

10. Logic, language and proof

LOGIC WORDS

"A and B" means both are true.

"A or B" means inclusive or: A, B, or both.

"Not A" means A is false.

TMUA tests precise reading of English.

IF / ONLY IF / IFF

"If A, then B" means $A \Rightarrow B$.

"A if B" means $B \Rightarrow A$.

"A only if B" means $A \Rightarrow B$.

"A iff B" means both $A \Rightarrow B$ and $B \Rightarrow A$.

CONVERSE AND CONTRAPOSITIVE

Statement: $P \Rightarrow Q$

Converse: $Q \Rightarrow P$, not equivalent in general.

Inverse: $\neg P \Rightarrow \neg Q$, not equivalent in general.

Contrapositive: $\neg Q \Rightarrow \neg P$, always equivalent.

NECESSARY AND SUFFICIENT

If $P \Rightarrow Q$:

P is **sufficient** for Q.

Q is **necessary** for P.

"Fire implies oxygen": oxygen is necessary for fire.

QUANTIFIERS

"For all" means every case.

"For some" means at least one case.

"There exists" means at least one example exists.

NEGATING QUANTIFIERS

$$\neg(\forall x, P(x)) \iff \exists x, \neg P(x)$$

$$\neg(\exists x, P(x)) \iff \forall x, \neg P(x)$$

"Not all" means "at least one not".

DE MORGAN

$$\neg(P \wedge Q) \iff \neg P \vee \neg Q$$

$$\neg(P \vee Q) \iff \neg P \wedge \neg Q$$

In English: not both = one or both fail.

TRUTH-TABLE FACTS

learning aid $P \Rightarrow Q$ is false only when P is true and Q is false.

A biconditional $P \iff Q$ is true when P and Q have the same truth value.

SET NOTATION

$\cup$: union, $\cap$: intersection, $\in$: element of

$\subset$: subset, $\emptyset$: empty set, A' : complement

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

COUNTEREXAMPLES

To disprove "for all x, P(x)", one counterexample is enough.

To disprove "there exists x such that P(x)", prove no case works.

DIRECT PROOF

Start from known assumptions and proceed logically:

$$A \Rightarrow B \Rightarrow C \Rightarrow \dots \Rightarrow Z$$

Each step must follow from earlier steps.

PROOF BY CASES

Split into exhaustive cases, e.g. even/odd, positive/negative/zero.

Every possible case must be covered.

PROOF BY CONTRADICTION

To prove S, assume $\neg S$.

Derive an impossibility.

Conclude S must be true.

PROOF BY CONTRAPOSITIVE

To prove $P \Rightarrow Q$, prove:

$$\neg Q \Rightarrow \neg P$$

Often easier when Q contains a negative condition.

INDUCTION

extra Base case: prove $P(1)$.

extra Inductive step: assume $P(k)$, prove $P(k+1)$.

extra Conclude $P(n)$ for all required integers n.

COMMON PROOF ERRORS

Invalid: $ab = ac \Rightarrow b = c$ unless $a \neq 0$.

Invalid: $\sin A = \sin B \Rightarrow A = B$ without checking other angle solutions.

Invalid: proving examples and claiming the general result.

PAPER 2 PANIC CHECKLIST

- Underline "must", "could", "only if", "for all", "some", "not".
- Try small cases: 0, 1, 2, even/odd, positive/negative.
- Look for a counterexample before doing heavy algebra.
- For implications, test the contrapositive and the converse separately.
- No negative marking: eliminate, guess, flag, move.